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Characteristics of the plasma sheath in helium discharge within dielectric tubes

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Abstract

To understand the characteristics of the plasma sheath within small tubes, a 2D numerical model of He discharge within dielectric tubes is developed. During plasma propagation for a tube diameter of 0.05 mm, the sheath thickness in the plasma head is almost equal to the tube radius. It decreases rapidly to several micrometers at an axial distance of 0.05 mm behind the plasma head, and then slightly increases and saturates at the axial position far behind the plasma head. A plasma-gas sheath surrounding the central plasma column is observed for a tube diameter equal to or greater than 0.8 mm. It is replaced by a plasma-wall sheath for smaller tubes. With the decrease in the tube diameter, the sheath thickness far behind the plasma head decreases while the ion flux increases significantly. However, when O_2 gas with a proportion of 2% is added, both the sheath thickness and ion flux decrease.

Keywords: plasma sheath, plasma modification, helium discharge, ion flux, plasma jet

(Some figures may appear in colour only in the online journal)

1. Introduction

Polymer tubes have been extensively applied in biomedicine, microelectronics and fluid transport due to their excellent features including high mouldability, electric insulation, optical transparency, excellent chemical resistance, light weight and good quality biocompatibility. However, the hydrophobic nature of polymer tubes with low surface free energy leads to poor adhesive properties and, therefore, limitation of applications where a good contact with other materials or cells is required. Previous works have shown that an increase in the surface free energy and the surface roughness can efficiently improve the adhesion strength of polymer tubes [1]. Compared to the traditional method of wet chemical treatment processes, plasma modification (including plasma etching, plasma-assisted deposition and plasma treatment) is an environmentally friendly method that can improve the surface hydrophilicity in nanometer to a few micrometers scale without changing the bulk properties.

Compared to the low-pressure plasma operated in a vacuum system, atmospheric-pressure non-thermal plasma has the advantage of low cost, low gas temperature, all-dry processing and high efficiency. Therefore, atmospheric-pressure non-thermal plasma sources (such as capacitively coupled discharge, pulsed corona discharge and pulsed-DC glow discharge) have attracted much attention in plasma modification of the inner surface of polymer tubes. The disadvantage of these plasma sources is the complicated procedure of setting up the electrode precisely on the polymer tubes. High breakdown voltage is required and to which the operators are exposed [2–6]. Non-thermal atmospheric-pressure plasma jets, rich in reactive chemical species at room temperature, show advanced surface modification, sterilization and decontamination applications [7–13]. Chen *et al* developed a transferred atmospheric-pressure plasma inside polytetrafluoroethylene (PTFE) tubes ignited by an argon plasma jet [14]. Xu *et al* reported the modification of the tube

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inner surface by a He/O_2 microplasma jet [1]. Onyshchenko *et al* investigated plasma injection into a polymeric tube by a plasma jet [15]. Wang *et al* demonstrated significant improvement in the wettability of the tube's inner walls by a flexible microplasma jet injected into a PTFE tube [16].

Plasma sheath, which determines the ion energy and ion flux on the surface, plays an essential role in modifying the inner surface of the polymer tubes. In addition, it dramatically impacts the dynamics of the plasma propagation within tubes. Lietz *et al* reported the numerical modeling of He/air plasma jets and showed that a plasma sheath forms near the inner surface of the dielectric tube, where the tube walls are negatively charged [17]. Similar results have also been reported by Sharma *et al* [18]. Furthermore, the presence of the plasma sheath is closely related to the structure of the plasma volume in the capillary discharge. Liu *et al* reported that the plasma sheath causes the plasma to shrink inside the tube [19]. For the He plasma jets inside tubes with multi-ring electrodes, Li *et al* concluded that the formation of the plasma sheath near the inner surface of the tube walls should be responsible for the presence of the ring-shaped structure of the plasma volume inside the tubes. The sheath depends on the radial electric field strength that supports the gas ionization near the inner walls [20]. In addition, Ning *et al* presented numerical modeling of He plasma jets with different tube diameters, and found that the shape of the plasma volume transits from a ring-shaped mode to a solid-shaped mode for a sheath thickness close to the tube radius [21]. The works mentioned above show that the formation of the plasma sheath near the inner surface of the tube walls has a great impact on the plasma propagation inside the tubes. Nevertheless, the spatial-temporal evolution of the plasma sheath near the inner walls in the pulsed plasma jets is not well understood. Therefore, based on the 2D modeling of He or He/O_2 discharges inside tubes, this study quantitatively investigates the characteristics of the plasma sheath and how it is influenced by the tube diameter and the addition of O_2 gas.

2. The model

2.1. Simulation configuration

Figure 1 shows the 2D axis-symmetric geometry. A quartz tube is filled with He at atmospheric pressure and the background gas temperature is 293.15 K. In atmospheric-pressure plasma jets (APPJs), the ion temperature is close to the gas temperature at atmospheric pressure. The inner diameter of the tube is in the range of 0.05–1 mm, and the length of the tube is 5 mm. The relative dielectric constant of the tube is 4.3. It should be pointed out that a quartz glass (fused silica), instead of borosilicate glass, is used in the simulation and experiment. The breakdown field of the quartz glass is about 4.7–6.7 MV cm^{-1} [22]. The simulation of a static electric field shows that the maximum electric field near the needle tip is much lower than the breakdown field of the quartz glass. In addition, no breakdown of the dielectric tube made of quartz

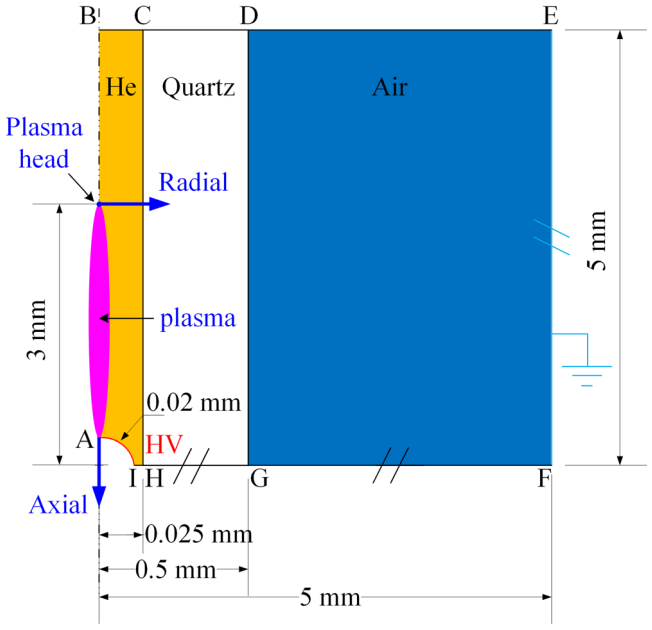


Figure 1. Schematic of the simulation configuration.

glass (bought from Hilgenberg GmbH) was observed in our previous experiment [23]. Furthermore, a DC-pulsed voltage (instead of DC voltage) with a pulse rise time of 400 ns is applied to the needle, which further reduces the possibility of electrical breakdown of the quartz tube at room temperature [24]. Therefore, electrical breakdown of the quartz tube is unlikely to occur.

A needle with a radius of 0.02 mm is inserted into the quartz tube. The amplitude and rise time of the applied pulsed voltage are 18 kV and 400 ns, respectively. For fast convergence, an air layer surrounding the tube is added, and its outer surface is grounded. As the radial length of the calculation zone increases to 10 mm, the Laplace electric field inside the tube remains unchanged. The plasma head is defined as the zero position of the axial distance for the following study. Once the excited species reach the inner surface of the tube wall, they are quenched to the ground state. The secondary electron emission coefficient of the tube wall is 0.1, which roughly includes the contribution of the photoemission, secondary emission of electrons by impact of metastable molecules and field emission [25]. The surface reactions are set at the inner surface of the tube, where excited or ionic species transit back to their ground states. The wall reaction probability of neutral species is 1. Considering the charges produced by the natural radiation photoionization, the initial densities of excited species are set at 25 cm^{-3}.

In air photoionization, photons are emitted by $N_2(b^1\Pi)$ and $N_2(b^1\Sigma)$ with a wavelength of 98–102 nm and absorbed by O_2 molecules, where the photoionization coefficient is related to the electric field [26]. On the other hand, in pure He gas, a one-step photoionization scheme cannot exist in the He discharges. In He/O_2 gas mixture, photoionization may present, where the photons are emitted from the resonance radiation from He(21 P) and absorbed by O_2 molecules. However, there are no available data on the free electron

generation by He photoionization at present. With and without the consideration of air photoionization in He plasma jets, Breden *et al* [27] found that air photoionization is not an essential process for streamer propagation. In addition, as 1% O_2 is added to the He gas, the velocity of the streamer propagation slightly increases, and the air photoionization rate is several orders of magnitude lower than the electron impact ionization rate and the Penning ionization rate [28, 29]. Furthermore, in the non-thermal He plasma jet driven by AC or pulsed-DC voltage with frequency of several kHz, the background electrons left from previous discharges are accumulated, which can supply seed electrons for the plasma propagation. Our previous experimental study [30] found that the background electron density can be as high as $10^9\,\mathrm{cm}^{-3}$, which is significantly higher than the electron density (10^6–$10^8\ \mathrm{cm}^{-3}$) produced from the air photoionization [27]. In other words, photoionization plays a minor role in the propagation of the He plasma plume for kHz repetitive discharge. Therefore, for the simulation of He plasma jet in previous works [19, 31, 32], background electron density of 10^7–$10^{10}\ \mathrm{cm}^{-3}$ is usually assumed. In this study, a background electron density of $10^9\,\mathrm{cm}^{-3}$ is assumed, and the He photoionization is not taken into account in the model.

Runaway electrons are usually observed in fast pulsed discharge with high voltage up to several hundred kV and a voltage rise time of less than 10 ns [33, 34]. In this study, the generation of the runaway electrons and its effects on the He plasma confined inside small tubes are analyzed below. First, because the electric field in front of the plasma head at the tube diameter of less than 0.8 mm is greater than the critical electric field strength for He (50–100 $\mathrm{kV\,cm}^{-1}$), the runaway electrons can be generated in the He capillary discharge. According to [35–37], theoretical and numerical simulation results showed that the production rate of the runaway electrons is greatly dependent on the electric field, maximum simulated energy of electrons, and gas species. In our study, the spatial distribution of the electric field in front of the plasma head inside the small tube is extremely non-uniform and fast time-varying. The maximum electric field strength in front of the plasma head reaches as high as $330\,\mathrm{kV\,cm}^{-1}$ (obtained from the numerical results in the following section 3.1) on the axis of symmetry for a tube diameter of 0.05 mm. The characteristic space scale of the electric field gradient (λ_E) in this plasma is about 0.025 mm. This means that the maximum energy of electrons in this plasma should be significantly less than 1 keV, which is different to the term 'runaway electron' in the literature that an electron with energy greater than 1 keV is usually considered to be a runaway one [36, 38, 39]. Therefore, these 'runaway electrons' generated in the He microplasma in our study are probably low-energy electrons. Moreover, numerical results computed by the Monte Carlo method showed that the runaway rate (ν_run) at a high electric field of $300\,\mathrm{kV\,cm}^{-1}$ is $7 \times 10^{11}\,\mathrm{s}^{-1}$ with the maximum simulated energy of electrons ε_max of 1 keV [36], where the average length l_he at which the electrons gain the energy of 1 keV is about 0.038 mm. However, because the λ_E in our study is much smaller than l_he of 0.038 mm with electron energy of 1 keV, this further increases the probability of the generation of low-energy electrons (instead of high-energy electrons) in our study. Second, Bochkov *et al* [36] further pointed out that to estimate the source of runaway electrons generated in the He plasma by the product of $\nu_\mathrm{run} \times n_\mathrm{le}$, the condition $l_\mathrm{he} \ll \lambda_\mathrm{E}$ has to be fulfilled first. Unfortunately, the l_he increases with the increase in the energy ε_max, and it reaches as large as 0.01 mm at an electric field of $300\,\mathrm{kV\,cm}^{-1}$ and an energy ε_max of 0.25 keV [36], which is comparable to the λ_E in our study. Therefore, the estimation of the source of runaway electrons by the product of $\nu_\mathrm{run} \times n_\mathrm{le}$ is not valid for the energy ε_max greater than 0.25 keV, at least in the He microplasma simulation. More importantly, the experimental data of the source of the low-energy runaway electrons in the He plasma are not available at present. In addition, with pulse voltage of a rise time of 400 ns and amplitude of several kV during the discharge, the He microplasma is generated inside small tubes and without contact with the grounded electrode, which is different to the He plasma generated between parallel-plate electrodes or pin-to-plate electrodes in traditional studies of runaway electrons [34, 36]. Therefore, the estimation of the source of the runaway electrons in the He microplasma in our study is significantly challenging. Third, the slow propagation velocity (10^5–$10^6\,\mathrm{m\,s}^{-1}$), the low-voltage growth rate ($4.5 \times 10^{10}\,\mathrm{V\,s}^{-1}$) and the small discharge current (<0.01 A) indicate that the propagation of the microplasma inside the small tubes behaves like a slow ionization wave, in which the seed electrons are mainly contributed by the electron emission from the inner tube surface (rather than runaway electrons), as reported by Akishev *et al* [40]. Therefore, the effects of the runaway electrons in this kind of slow ionization wave might play a minor role. Based on the aforementioned analysis, although low-energy runaway electrons are probably generated in the He microplasma confined inside small tubes, the effects of these runaway electrons on the He microplasma inside quartz tubes remain unknown at present. This is not considered in the plasma model and we hope to investigate it in a future study.

2.2. Plasma governing equations

The electric field ($\boldsymbol{E}$) is derived from the Poisson equation:

$$\boldsymbol{E} = -\nabla\varphi,\ -\nabla\varepsilon_0\varepsilon_\mathrm{r}\nabla\varphi = \rho, \tag{1}$$

where φ is the potential, ε_0 and ε_r are the vacuum dielectric constant and the relative dielectric constant, respectively; ρ is the volume charge density, given by,

$$\rho = q\left(\sum_{k=1}^{N} Z_k n_k - n_\mathrm{e}\right), \tag{2}$$

where q is the elementary charge, Z_k and n_k are the charge quantity and the number density of the kth charged species, respectively. The number densities of the species are calculated from the continuity equations:

$$\frac{\partial n_k}{\partial t} + \nabla\boldsymbol{\Gamma}_k = R_k, \tag{3}$$

where R_k is the source term including the production and the

Table 1. Boundary conditions of the simulation model.

	AB	CH	AI	IH/BC	EF	FG/DE	GH/CD
n_e	$\frac{\partial n_e}{\partial r}=0$	equation (6)	equation (6)	$\frac{\partial n_e}{\partial z}=0$	$\frac{\partial n_e}{\partial r}=0$	$\frac{\partial n_e}{\partial z}=0$	/
n_i	$\frac{\partial n_i}{\partial r}=0$	equation (7)	equation (7)	$\frac{\partial n_i}{\partial z}=0$	$\frac{\partial n_i}{\partial r}=0$	$\frac{\partial n_i}{\partial z}=0$	/
n_m	$\frac{\partial n_m}{\partial r}=0$	equation (8)	equation (8)	$\frac{\partial n_m}{\partial z}=0$	$\frac{\partial n_m}{\partial r}=0$	$\frac{\partial n_m}{\partial z}=0$	/
n_ε	$\frac{\partial n_\varepsilon}{\partial r}=0$	equation (9)	equation (9)	$\frac{\partial n_\varepsilon}{\partial z}=0$	$\frac{\partial n_\varepsilon}{\partial r}=0$	$\frac{\partial n_\varepsilon}{\partial z}=0$	/
φ	$\frac{\partial \varphi}{\partial r}=0$	equation (11)	Applied Voltage	$\frac{\partial \varphi}{\partial z}=0$	0	$\frac{\partial \varphi}{\partial z}=0$	$\frac{\partial \varphi}{\partial z}=0$

r is the radial direction; z is the axial direction.

loss of the kth species. $\boldsymbol{\Gamma}_k$ is the flux term of the kth species, which is described by the drift-diffusion approximation [19]:

$$\boldsymbol{\Gamma}_k = -\mu_k n_k \nabla\varphi - D_k \nabla n_k, \tag{4}$$

where μ_k and D_k are the charge mobility and the diffusion coefficient, respectively. The electron temperature is obtained from the electron energy conservation equation, described as [19],

$$\frac{\partial \overline{\varepsilon_e}}{\partial t} + \nabla\left(\frac{5}{3}\varepsilon\boldsymbol{\Gamma}_e - \frac{5}{3}n_e D_e \nabla\varepsilon\right) = e\boldsymbol{\Gamma}_e\cdot\nabla\varphi - e\sum_i \Delta E_i^e r_i - \frac{3}{2}k_b n_e \frac{2m_e}{m_{kb}}(T_e - T_g)\nu_{e,kb}, \tag{5}$$

where the total electron energy is approximately equal to the mean electron energy $\overline{\varepsilon_e} \approx \frac{3}{2}k_b n_e T_e$. ΔE_i^e is the energy loss from one inelastic collision event involving a reaction i with a rate of progress given by r_i. m_e and m_{kb} are the electron mass and gas species mass, respectively. $\nu_{e,kb}$ is the collision frequency for momentum transfer in elastic collisional events. T_e and T_g are the electron and background gas temperature, respectively. k_b is the Boltzmann constant.

2.3. Plasma boundary conditions

Table 1 shows the boundary conditions of the plasma model. The boundary conditions for the flux of particles and the electron energy density are [41],

$$\boldsymbol{\Gamma}_e\cdot\boldsymbol{n} = \frac{1}{4}n_e\sqrt{\frac{8k_bT_e}{\pi m_e}} - \alpha_s\sum_i \gamma_i(\boldsymbol{\Gamma}_i\cdot\boldsymbol{n}) + \alpha_s'\mu_e n_e \boldsymbol{E}, \tag{6}$$

$$\boldsymbol{\Gamma}_i\cdot\boldsymbol{n} = \frac{1}{4}n_i\sqrt{\frac{8k_bT_i}{\pi m_i}} + \alpha_s\mu_i n_i E, \tag{7}$$

$$\boldsymbol{\Gamma}_m\cdot\boldsymbol{n} = \frac{2\gamma_m}{2-\gamma_m}\frac{1}{4}n_m\sqrt{\frac{8k_bT_g}{\pi m_m}}, \tag{8}$$

$$\boldsymbol{\Gamma}_\varepsilon\cdot\boldsymbol{n} = \frac{1}{2}n_\varepsilon\sqrt{\frac{8k_bT_e}{\pi m_e}} - \alpha_s 2k_bT_e\sum_i \gamma_i(\boldsymbol{\Gamma}_i\cdot\boldsymbol{n}), \tag{9}$$

where e, i, m, and ε are electrons, ions, metastables and electron energy, respectively. γ_i is the secondary electron emission coefficient, and γ_m is the wall reaction probability of neutral species. $\boldsymbol{n}$ is the unit normal vector pointing towards the surface, and α_s and α_s' are switching functions depending on $\boldsymbol{E}\cdot\boldsymbol{n}$.

$$\alpha_s = \begin{cases} 1 & (\boldsymbol{E}\cdot\boldsymbol{n} \geqslant 0) \\ 0 & (\boldsymbol{E}\cdot\boldsymbol{n} < 0), \end{cases} \quad \alpha_s' = \begin{cases} 0 & (\boldsymbol{E}\cdot\boldsymbol{n} \geqslant 0) \\ 1 & (\boldsymbol{E}\cdot\boldsymbol{n} < 0). \end{cases} \tag{10}$$

The net charge density accumulated on the dielectric surface (σ_s) was calculated by Gauss's law:

$$(\boldsymbol{D}_2 - \boldsymbol{D}_1)\cdot\boldsymbol{n} = \sigma_s, \quad \frac{d\sigma_s}{dt} = \boldsymbol{J}_i\cdot\boldsymbol{n} + \boldsymbol{J}_e\cdot\boldsymbol{n}, \tag{11}$$

where $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ are the electric displacement vectors, and $\boldsymbol{J}_e$ and $\boldsymbol{J}_i$ are the electron current density and ion current density, respectively.

2.4. Plasma chemistry

Sixteen species are employed in the plasma model for e, He, He^+, He^*, He_2^*, He_2^+, O_2, O_2^+, O, O^-, O_2^-, O_2(A1), O_2(B1), O_2(EXC), O(1D) and O(1S). Thirty-four reactions are included in table 2 [19, 42]. The reaction rates of the direct collisions between electrons and atoms or molecules are calculated by Bolsig + [42]. The Penning ionization of O_2 by He metastable is taken into account. Reaction rates are presented in the form of Arrhenius expression $AT_e^B\exp(-C/T_e)$. It should be pointed out that the O_2-related reactions are turned off when only He discharge is considered.

2.5. Solution settings

Plasma Module in COMSOL Multiphysics software is applied to solve the above-mentioned equations [43]. In order to ensure the correctness and accuracy of the solution, we use triangular finite elements to divide the mesh. The He discharge region is divided into finer grids where the maximum value of the mesh size is 0.001 mm. The regions of quartz glass and air are divided automatically, where the maximum value of the mesh size is about 0.5 mm. The computation domain of the plasma model is divided into a total of 380 000 elements with 1 100 000 degrees of freedom. The maximum time step is automatically set at 10^{-13} s. The parallel sparse direct solver is chosen due to its excellent robustness.

Table 2. He-O_2 chemical reactions used in the model [19, 42].

Number	Reaction	Reaction rates		
		A	*B*	*C*
He chemistry				
R1	$e + He \rightarrow e + He$	BOLSIG+		
R2	$e + He \rightarrow e + He^*$	BOLSIG+		
R3	$e + He \rightarrow 2e + He^+$	BOLSIG+		
R4	$e + He^* \rightarrow 2e + He^+$	4.661×10^{-16}	0.6	4.78
R5	$2He^* \rightarrow e + He + He^+$	4.5×10^{-16}	0	0
R6	$He^* + 2He \rightarrow He + He_2^*$	1.3×10^{-45}	0	0
R7	$e + He_2^* \rightarrow 2e + He_2^+$	1.268×10^{-18}	0.71	3.4
R8	$e + He^+ + He \rightarrow He^* + He$	6.66×10^{-42}	−2	0
R9	$He^+ + 2He \rightarrow He + He_2^+$	1.0×10^{-43}	0	0
R10	$e + He^+ \rightarrow He^*$	6.76×10^{-19}	−0.5	0
R11	$2e + He^+ \rightarrow e + He^*$	6.186×10^{-39}	−4.4	0
R12	$e + He_2^+ \rightarrow He^* + He$	5.386×10^{-13}	−0.5	0
R13	$2e + He_2^+ \rightarrow e + He_2^*$	2.8×10^{-32}	0	0
R14	$e + He_2^+ + He \rightarrow He^* + 2He$	3.5×10^{-39}	0	0
R15	$e + He_2^+ + He \rightarrow He_2^* + He$	1.5×10^{-39}	0	0
R16	$2e + He_2^+ \rightarrow e + He^* + He$	2.8×10^{-32}	0	0
O_2 chemistry				
R17	$e + O_2 \rightarrow e + O_2$	BOLSIG+		
R18	$e + O_2 \rightarrow O + O^-$	BOLSIG+		
R19	$e + O_2 \rightarrow e + 2O$	BOLSIG+		
R20	$e + O_2 \rightarrow 2e + O_2^+$	BOLSIG+		
R21	$e + O_2 \rightarrow e + O_2(A1)$	BOLSIG+		
R22	$e + O_2 \rightarrow e + O_2(B1)$	BOLSIG+		
R23	$e + O_2 \rightarrow e + O_2(EXC)$	BOLSIG+		
R24	$e + O_2 \rightarrow e + O + O(1D)$	BOLSIG+		
R25	$e + O_2 \rightarrow e + O + O(1S)$	BOLSIG+		
R26	$e + O_2(A1) \rightarrow e + O_2$	BOLSIG+		
R27	$e + O_2(B1) \rightarrow e + O_2$	BOLSIG+		
R28	$e + O_2(EXC) \rightarrow e + O_2$	BOLSIG+		
R29	$O^- + O_2^+ \rightarrow O + O_2$	3.488×10^{-14}	−0.5	0
R30	$e + 2O_2 \rightarrow O_2^- + O_2$	5.17×10^{-43}	−1	0
R31	$O_2^- + O_2^+ \rightarrow +2O_2$	2.0×10^{-13}	0	0
He+O_2 chemistry				
R32	$He^* + O_2 \rightarrow He + O_2^+ + e$	2.6×10^{-16}	0	0
R33	$He_2^* + O_2 \rightarrow 2He + O_2^+ + e$	3.6×10^{-16}	0	0
R34	$He_2^+ + O_2^- \rightarrow 2He + O_2$	1.0×10^{-13}	0	0

The units of reaction rates are $m^3\ s^{-1}$ for two-body reactions and $m^6\ s^{-1}$ for three-body reactions.

3. Results and discussion

3.1. Plasma sheath characteristics

Experimental He discharges within tubes with diameters of 0.4 and 0.05 mm are implemented to verify the simulation results. The applied voltage is the same as that in the model. A similar experimental setup is presented in [23, 44]. The pulsed-DC voltage with an amplitude of 18 kV and pulse rise time of 400 ns is applied to the high-voltage electrode. A fast intensified charged-coupled device (ICCD) camera (Andor DH334T-18U-03) is used to capture the dynamics of the He discharge within the tubes. The exposure time of the ICCD is 3 ns. In figure 2, the simulation time is 134 and 164 ns for tube diameters of 0.4 and 0.05 mm, respectively. The experimental time corresponding to the rise in the voltage pulse is 109 and 103 ns for tube diameters of 0.4 and 0.05 mm, respectively. When the tube diameter is 0.4 mm, ICCD images show that the plasma volume propagates forwards like a discrete 'plasma bullet'. The dark channel occurs between the 'plasma bullet' and the needle tip. However, for the tube diameter of 0.05 mm, a continuous plasma column is presented, which is always connected to the needle tip. This mode transition has been reported in our previous work [45]. Because the pixel values of the ICCD images are derived from the photons that are mainly emitted from the transition from He excited states to the ground state, the simulated results of the distribution of He excited states are used to characterize the shape of the He plasma volume. In figure 2, the maximum density of the He excited state in the middle of

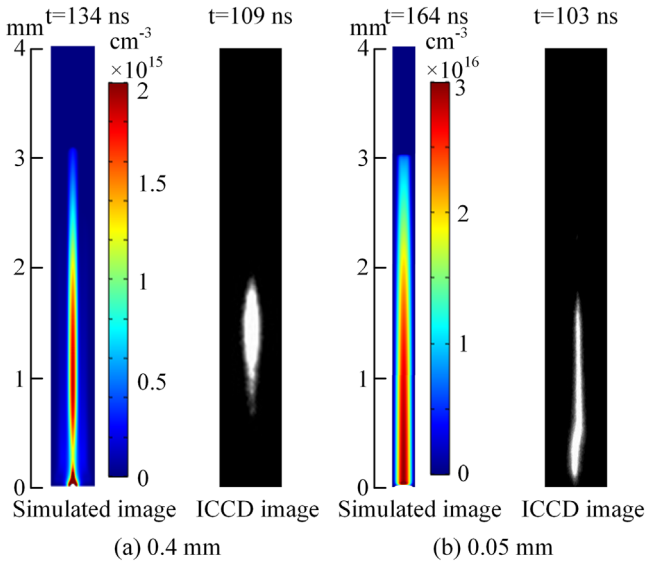


Figure 2. Simulated 2D distribution of the He excited states and the ICCD images of the He discharge. (a) 0.40 mm; (b) 0.05 mm. Labeled time in each graph corresponds to the rise in the voltage pulse.

the plasma channel is close to $2 \times 10^{15}\,\mathrm{cm}^{-3}$ for a tube diameter of 0.4 mm. However, the density of the He excited state near the needle is apparently lower than that in the middle of the plasma channel, which means that a dark region between the plasma volume and the needle tip presents, similar to the 'plasma bullet'. On the other hand, the density of the He excited state near the needle tip reaches as high as $2 \times 10^{15}\,\mathrm{cm}^{-3}$ for the tube of 0.05 mm, and decreases along the plasma channel monotonically, which is consistent with the experimental sword-shaped plasma column that strong light radiates near the needle tips. Therefore, the simulated results are in good agreement with the experimental results. In addition, the density of the He excited states in the case of 0.4 mm is much lower than that for the case of 0.05 mm.

The radial distributions of ion density, the electron density (n_e), the electric field strength, and the electron temperature at different axial positions around the head of the plasma volume are plotted in figure 3. The tube diameter is set at 0.05 mm. The axial position of the plasma head is defined as the position where the n_e is $1/e$ times the maximum n_e. Figure 3 shows that the n_e is much lower than the ion density in front of the plasma head, where the electric field is greater than $300\,\mathrm{kV\,cm}^{-1}$ and the electron temperature is over 60 eV. In the plasma head, the n_e is almost equal to the ion density at the axis but much lower than the ion density near the tube wall. A sharp decrease in the electric field strength from the tube wall to the axis is observed. The electron temperature first increases and then decreases from the wall to the axis, where the maximum electron temperature reaches 54 eV at a radial distance of 0.02 mm. At the axial positions far behind the head of the plasma volume, as the position where the n_e is equal to the ion density moves towards the tube wall, the electric field becomes weaker, and the electron temperature drops. For the axial position behind the plasma head at 0.05 mm, the electric field strength on the tube wall is as low as $9\,\mathrm{kV\,cm}^{-1}$, and the peak value of the electron temperature drops to 3.68 eV. Therefore, characterized by a rapid decrease in the electric field and a relatively high ion density, a typical plasma sheath between the plasma column and the tube wall is observed during the propagation of the plasma volume.

Based on the radial distributions of the charge density and the electric field strength in figure 3, the thickness of the plasma-wall sheath is defined as the minimum radial distance between the tube wall and the radial position where the n_e is equal to the ion density. Figure 4 shows the spatial distributions of the sheath thickness at different times. At the head of the plasma volume, the sheath thickness is equal to the inner radius of the tube. At 128 ns, when the axial distance increases, the sheath thickness drops fast down to 3.7×10^{-3} mm at the axial distance of 0.05 mm, and then it increases and saturates at 8.3×10^{-3} mm. For axial distance greater than 0.05 mm, the sheath thickness decreases slightly as the delay time increases from 128 to 164 ns. In addition, the Debye length in the plasma at the axial distance of 0.05 mm is calculated by the equation $\lambda_D = \sqrt{\varepsilon_0 kT_e/n_e q^2}$, where the electron temperature and the n_e are 3.61 eV and $1.8 \times 10^{15}\,\mathrm{cm}^{-3}$, respectively. The Debye length is estimated at 3.3×10^{-4} mm, which is ten times less than the minimum thickness of the plasma sheath, similar to the relationship between the Debye length and the sheath thickness obtained in DBDs at high pressure [46, 47].

3.2. Effects of the tube diameter

According to the above-mentioned results, the thickness of the plasma-wall sheath behind the head of the plasma volume is about 6×10^{-3}–8.3×10^{-3} mm, which is about 1/3–1/4 times the tube radius of 0.025 mm. Since the plasma is constrained inside the tubes, the diameter of the tubes may have a significant impact on the characteristics of the plasma sheath.

Figure 5 shows the 2D distributions of the n_e for different tube diameters. The n_e around the axis is about $1.3 \times 10^{13}\,\mathrm{cm}^{-3}$ for a tube diameter of 1 mm. The plasma column is mainly concentrated at the axis, and the diameter of the plasma column is approximately 1/3 times the tube diameter. For the tube diameter reduced to 0.2 mm, a plasma layer with relatively lower n_e surrounding the central plasmas column is observed, which has not previously been reported. The central plasma column disappears unexpectedly at a tube diameter of 0.1 mm, and only the plasma layer near the inner wall is presented, resulting in the observation of a typical ring-shaped mode with high n_e near the inner wall and low n_e in the center of the plasma column. However, for a tube diameter as small as 0.05 mm, a solid-shaped plasma column is presented, and the maximum n_e reaches as high as $4 \times 10^{15}\,\mathrm{cm}^{-3}$. Using the Stark broadening method of H_α line at a wavelength of 656.3 nm, we experimentally measured the electron density of the He microplasma confined inside the tube of diameter of 0.05 mm. The Stark broadening width of H_α line is 0.128 nm, which yields electron density of $4.2 \times 10^{15}\,\mathrm{cm}^{-3}$. It is well consistent with the simulation results. In other words, it further enhances the correctness of the plasma model. In

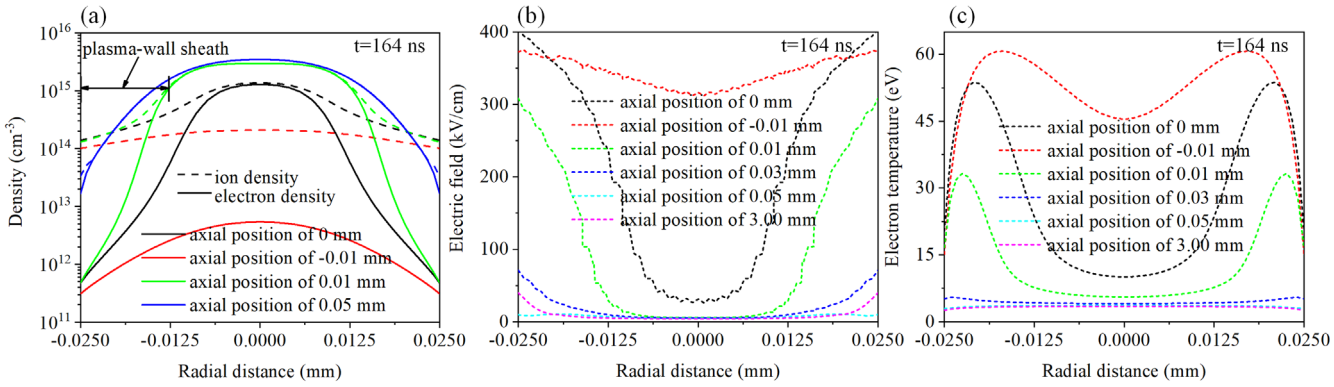


Figure 3. Radial distributions of (a) the charge densities, (b) the electric field strength, and (c) the electron temperature at different axial positions near the head of the plasma volume at 164 ns. Diameter of the tube is 0.05 mm. Dashed and solid lines in figure 3(a) are the profiles of the ion density and n_e, respectively. Axial position of 0 mm corresponds to the head of the plasma volume. Labeled time in each graph corresponds to the rise in the voltage pulse.

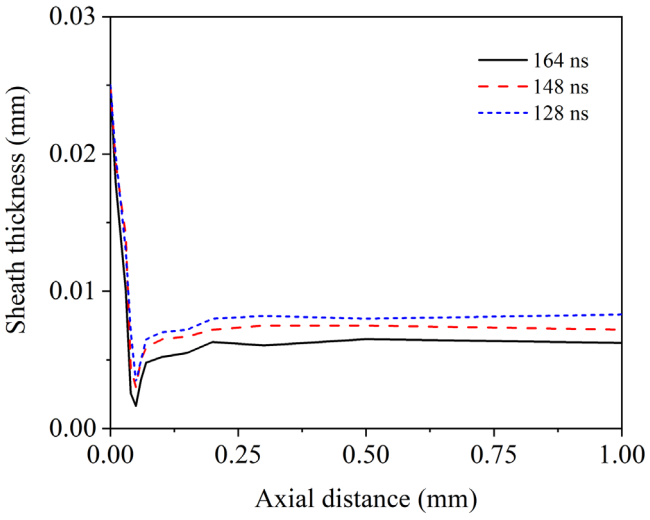


Figure 4. Sheath thickness of the He plasma at different delay times. Diameter of the tube is 0.05 mm. Zero position represents the head of the plasma volume. Labeled time corresponds to the rise in the voltage pulse.

addition, the maximum electron density of the He microplasma is four orders of magnitude lower than the density of the background He gas, which indicates that the background He gas is partially ionized, instead of fully ionized.

To understand the mode transition of the He discharge, 2D distributions of the electric field are presented in figure 6. In addition to the maximum electric field of 120 kV cm^{-1} in front of the plasma volume, the electric field near the inner wall reaches as high as 80 kV cm^{-1} in the case of 0.3 mm. This causes ionization of the He gas and, therefore, the presence of the extra plasma layer near the inner wall. At 0.2 mm, the maximum electric field strength near the wall is close to the electric field strength in front of the plasma volume. Moreover, a maximum electric field of 300 kV cm^{-1} presents near the inner surface of the tube wall, and the central electric field becomes relatively weak in the case of 0.1 mm. This is probably due to the shielding effects of the plasma layer near the tube wall on the central plasma column, which causes the disappearance of the plasma column at the axis and the formation of ring-shaped plasma, as shown in figure 5. When the tube diameter is further reduced to 0.05 mm, the maximum electric field increases to 400 kV cm^{-1}, and a solid-shaped plasma column is obtained due to the space limitation.

For the tube diameters of 0.05 and 0.1 mm, the maximum electric field strength occurs near the tube wall, which corresponds to the boundary of the plasma-wall sheath. The maximum electric field strengths on the axis of symmetry for the tube diameters of 0.05 and 0.1 mm are 330 and 180 kV cm^{-1}, respectively. The electric field strength for the tube diameter of 0.1 mm is close to the numerical results reported by Ning *et al* [21], where the maximum electric field on the axis of symmetry reaches about 204 kV cm^{-1} in the He microplasma confined inside a capillary with a diameter of 0.1 mm. In addition, Gazeli *et al* [48] presented a numerical modeling of a He plasma jet for a tube diameter of 0.25 mm. At the initial stage of the plasma propagation, a large field of ~100 kV cm^{-1} near the high-voltage electrode is observed, and then decreases over a time of ~0.1 μs to about 85 kV cm^{-1} with applied voltage of 1.7 kV. In our simulation, the amplitude and the rise time of the pulse voltage are 18 kV and 400 ns, respectively. Accordingly, the voltage rises to 1.7 kV at 96 ns when the maximum electric field reaches 62 kV cm^{-1} in the case of a tube diameter of 0.3 mm, which is close to the aforementioned results reported by Gazeli *et al*. Therefore, the results of the electric field strength for a tube diameter equal to or greater than 0.1 mm are in good agreement with other results. Moreover, according to the dependence of the electric field on the tube diameter in previous works [21, 45, 49, 50], when the tube diameter is further reduced to 0.05 mm, the maximum electric field strength in front of the plasma head will further increase. This means that the maximum electric field strength of 330 kV cm^{-1} on the axis of symmetry for the tube diameter of 0.05 mm is possible.

Jansky and Bourdon devoted a significant amount of time to the modeling of streamer propagation in tubes filled with He [51–54]. Jansky *et al* reported the simulation of a He discharge within thin tubes at atmospheric pressure [51]. The inner diameter of the tube is 1 mm. The maximum electron

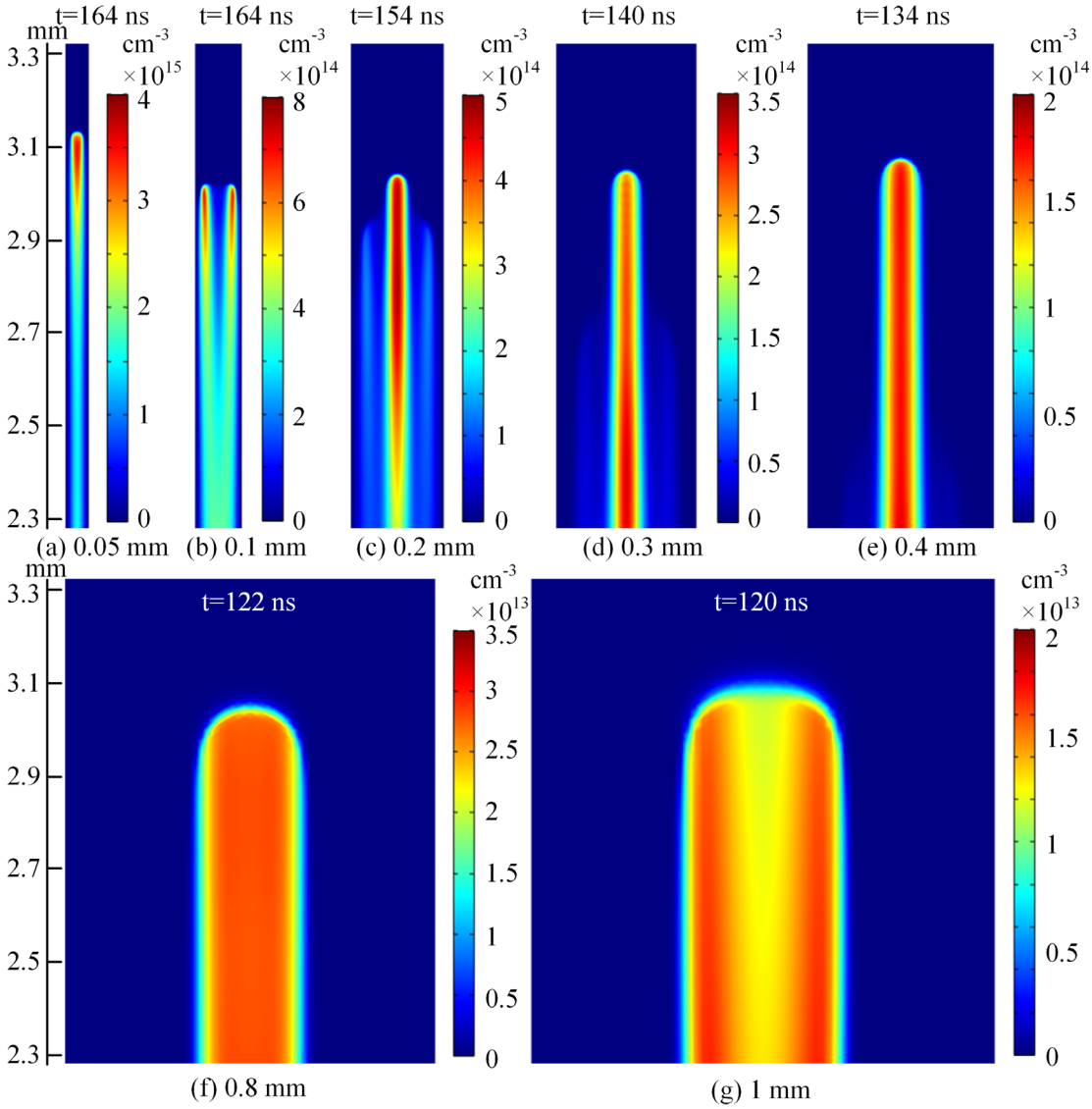


Figure 5. 2D distributions of the electron density (n_e) for different diameters of dielectric tubes. Labeled time in each graph corresponds to the rise in the voltage pulse.

density is about 10^{15} cm^{-3} and the maximum electric field in front of the plasma head is 100 kV cm^{-1}, which are much higher than the values in this study. This may be caused by the different electrode configurations and the applied voltages. In their work, a DC voltage of 6 kV is applied to the needle. It is much greater than the pulse voltage of 3 kV at the simulation time of 120 ns in this study. On the other hand, in addition to the high-voltage needle, a grounded ring electrode is wrapped around the outer surface of the thin tube, which also strongly enhances the electric field. Nevertheless, both the distribution patterns of the electron density and the electric field are similar to our study.

Figure 7 shows the 2D distributions of the electron temperature for different tube diameters. For a tube diameter of 1 mm, the maximum electron temperature in front of the plasma head reaches about 10 eV. The smaller the tube diameter, the higher the electron temperature will be. For the tube diameter of 0.05 mm, the maximum electron temperature in front of the plasma head (rather than in the plasma head) is about 60 eV. It should be noted that the electron temperature is non-uniformly distributed. In front of the plasma head, the volume of the electron cloud with the electron temperature over 50 eV is as small as 3.93×10^{-5} mm^3, the electric field strength is in the range of 280–360 kV cm^{-1} and the electron density is in the range of 5×10^{10}–2×10^{12} cm^{-3}.

With regard to the work by Ning *et al* [21], the electron temperatures in the case of the tube diameter of 0.1 mm are not presented. They reported that the maximum electron temperature is about 7 eV for a tube diameter of 2 mm and 4.5 eV for a tube diameter of 6 mm. In our simulation, the maximum electron temperature is about 10 eV for the tube diameter of 1 mm. Because a decrease in the tube diameter leads to an increase in plasma parameters, it is reasonable to believe that the simulation results of the electron temperature and the electric field in our case agreed well with the numerical results of Ning *et al.*

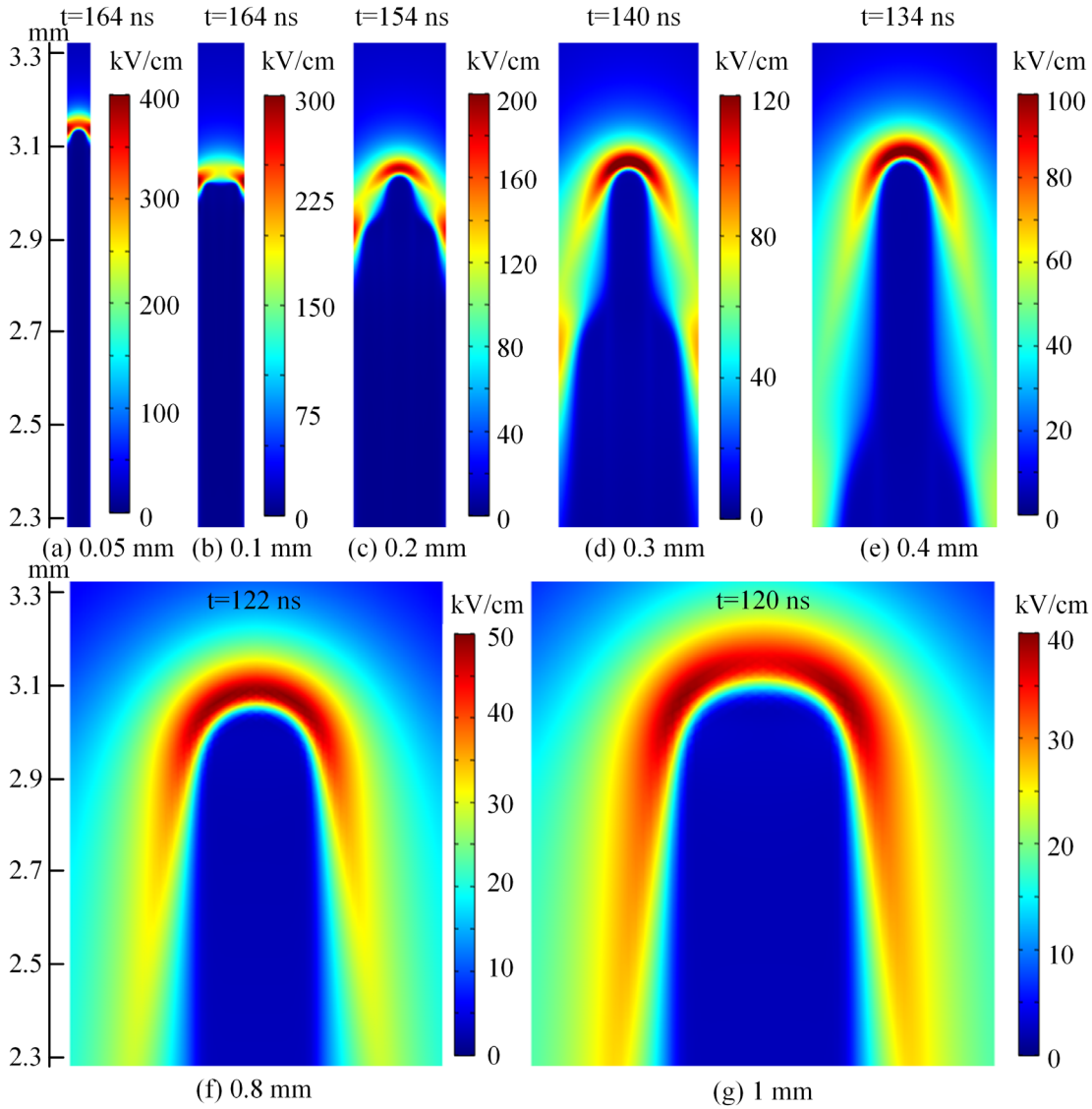


Figure 6. 2D distributions of the electric field for different tube diameters. Labeled time in each graph corresponds to the rise in the voltage pulse.

Figure 8 shows the radial distributions of the n_e and the electric field for different diameters of dielectric tubes. According to the results in figure 4, because the plasma sheath far behind the head of the plasma volume is almost independent of the axial distance, the plasma sheath at an axial distance of 1.5 mm is chosen for each case. For a tube diameter equal to or less than 0.1 mm, a plasma sheath is generated between the inner wall and the plasma column. For a tube diameter of 0.4 mm, a plateau of charge density profile between the plasma-wall sheath and the central plasma column is observed, where the ion density is equal to the n_e. The electric field strength on the inner wall is 27 kV cm^{-1}. On the other hand, in the case of 1 mm, a similar plateau of ion density profile presents close to the central plasma column, but the n_e is much lower than the ion density. As the axial distance increases, the electric field strength increases linearly first, and then reaches a maximum of 23 kV cm^{-1} at an axial distance of 0.375 mm, and finally decreases. This indicates that the characteristics of the plasma sheath for the tube diameter of 1 mm are largely different from those in other cases.

Because electrons are lighter and faster than ions, they propagate from the bulk plasma to the electrodes (or samples) at higher velocity than ions. Once the electrons are mostly lost at the boundary interface between the plasma and electrodes (or samples), a region near the surface of the electrodes (or samples) with dominant positive ions and neutrals will present, which is usually referred to as the plasma sheath. In this study, the He plasma is generated within the dielectric tubes with different diameters. For small tubes, a plasma-wall sheath is presented between the bulk plasma and the inner surface of the tube, which is characterized by the dominant positive ions and a sharp decrease in the electric field from the wall to the bulk plasma. The thickness of the plasma-wall sheath is defined as the distance between the inner surface of the tube wall and the radial position where the electron density is equal to the ion density. On the other hand, for large tubes, both the ion density and the electron density near the

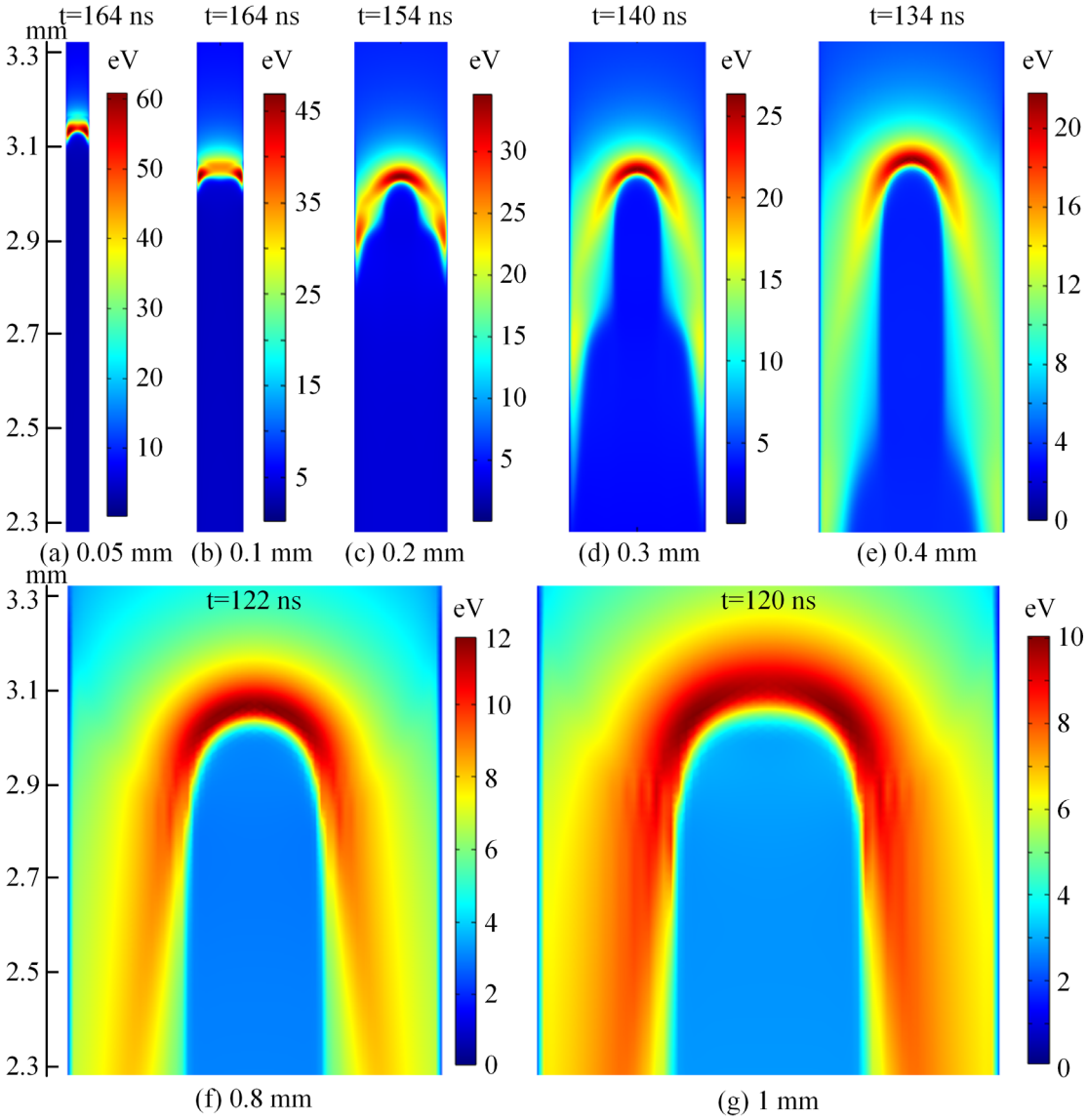


Figure 7. 2D distributions of the electron temperature for different tube diameters. Labeled time in each graph corresponds to the rise in the voltage pulse.

tube wall are extremely low and no sharp decrease in the electric field from the wall to the bulk plasma is observed, which represents that a gas layer, instead of the plasma-wall sheath, presents near the tube wall. Consequently, a region between the gas layer near the tube wall and the bulk plasma presents, where the ion density is significantly higher than the electron density, and a sharp decrease in the electric field from the boundary of the gas layer to the bulk plasma is observed. It can be clearly seen that the characteristics of this region are similar to those of the plasma-wall sheath. Therefore, for large tubes, the region between the gas layer near the tube wall and the bulk plasma is called the plasma-gas sheath.

To be consistent with the sheath characteristics in other cases, the thickness of the plasma-gas sheath in the case of 1 mm is defined as the distance between the blue dashed lines where the electric field increases monotonically with the increase in the axial distance. These phenomena can be explained as follows: the ionization process during the discharge is dominated by the electron impact ionization, and the charge diffusion and drifting processes to the inner surface of the tube wall are the primary loss processes of the electrons and ions. The diffusion and drifting processes of both the ions and electrons to the inner wall lead to the formation of the plasma sheath. The characteristic time of the diffusion and drifting processes can be calculated by $\tau_{\rm diff} = (R/2.4)^2/D$ and $\tau_{\rm drif} = R/(\mu E_{\rm r})$, respectively, where D is the diffusion coefficient of charges, μ is the charge mobility and $E_{\rm r}$ is the radial electric field. Table 3 shows the $\tau_{\rm diff}$ and $\tau_{\rm drif}$ at an axial distance of 1.5 mm for the tube diameters of 0.05, 0.1, 0.4 and 1 mm. The diffusion coefficient, the charge mobility and the electric field are derived from the simulation results in figure 7 and [55]. It is clearly shown that the $\tau_{\rm drif}$ is two orders of magnitude smaller than $\tau_{\rm diff}$. $\tau_{\rm drif-e}$ is also two orders of magnitude smaller than $\tau_{\rm drif-i}$, which is much faster than the duration of the plasma propagation. Therefore, the establishment of the plasma sheath is mainly determined by the

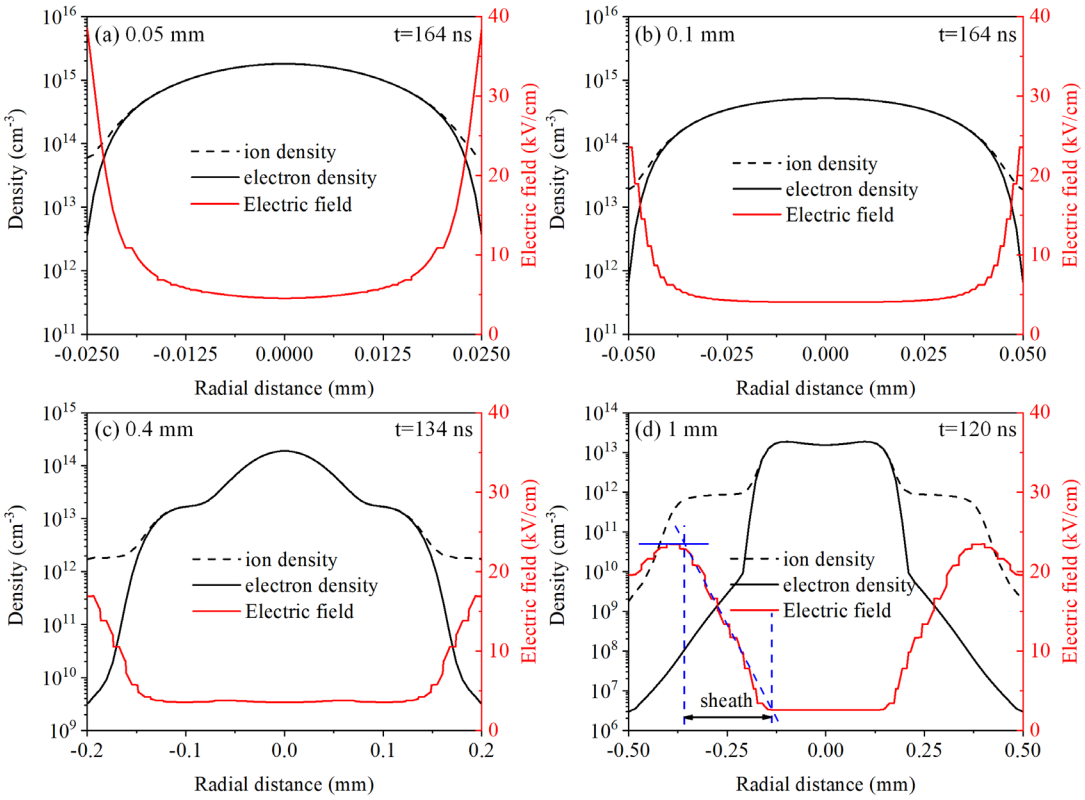


Figure 8. Radial distributions of the electric field and the charge density for different tube diameters at the axial position of 1.5 mm. Head of the plasma volume reaches the same axial position as that in figure 5. Labeled time in each graph corresponds to the rise in the voltage pulse.

Table 3. The estimated characteristic time of the electron diffusion, ion diffusion, electron drifting and ion drifting at the axial distance of 1.5 mm. The head of the plasma volume reaches the same axial position as that in figure 5.

Tube diameter (mm)	$\tau_{\mathrm{diff-e}}$ (ns)	$\tau_{\mathrm{diff-i}}$ (μs)	$\tau_{\mathrm{drif-e}}$ (ns)	$\tau_{\mathrm{drif-i}}$ (ns)
0.05	4.13	4.19	0.09	13.8
0.1	16.5	16.7	0.18	29.5
0.4	264	268	1.57	223
1	1650	1670	5.88	740

drifting process of the positive ions where $\tau_{\mathrm{drif-i}}$ is comparable to the duration time of the He discharge. The $\tau_{\mathrm{drif-i}}$ increases from 13.8 to 740 ns with the increase in the tube diameter from 0.05 to 1 mm. This indicates that the characteristic time of the formation of the plasma-wall sheath increases with the increase in the tube diameter. Moreover, the duration time for the propagation of the plasma volume is 164 ns in the case of 1 mm, which is smaller than the $\tau_{\mathrm{drif-i}}$. In other words, the formation of the plasma-wall sheath has not been accomplished during the propagation of the plasma volume, which leads to the observation of the plasma-gas sheath shown in figure 7. However, the $\tau_{\mathrm{drif-i}}$ is only 29.5 ns in the case of 0.1 mm, which is much less than the duration time for the plasma propagation. This indicates that the plasma-wall sheath can be established during the propagation of the plasma volume.

Figure 9 shows the axial distributions of the thickness of the plasma for different diameters of dielectric tubes. In the case of 1 mm, only plasma-gas sheath presents, and the sheath thickness increases monotonically with the increase in the axial distance. For the tube diameter of 0.4 mm, the sheath thickness rises first and then decreases after it reaches 0.143 mm at the axial distance of 0.5 mm. This corresponds to the sheath transition from the plasma-gas sheath to the plasma-wall sheath. For the tube diameters of 0.05 and 0.1 mm, only plasma-wall sheath presents, and it decreases rapidly at first, then increases slightly, and remains almost unchanged with the increase in the axial distance. These phenomena can be explained as follows: for the axial position far behind the head of the plasma volume, the ions are generated relatively earlier compared to those in the plasma head, and the ions drift radially farther away from the axis. Therefore, in the case of 1 mm, the greater the axial distance, the longer the time for ions to drift, and the thicker the plasma-gas sheath. However, for the tube diameter equal to or less than 0.1 mm, the drifting time of ions towards the tube wall is only tens of nanoseconds. The plasma-wall sheath far behind the head of the plasma volume has been well established during the propagation of the plasma volume. Therefore, the thickness of the plasma-wall sheath remains almost unchanged with the increase in the axial distance. At the head position of the plasma volume, the ions are generated around the axis and do not have enough time to reach the inner wall, leading to a relatively thicker plasma-wall sheath.

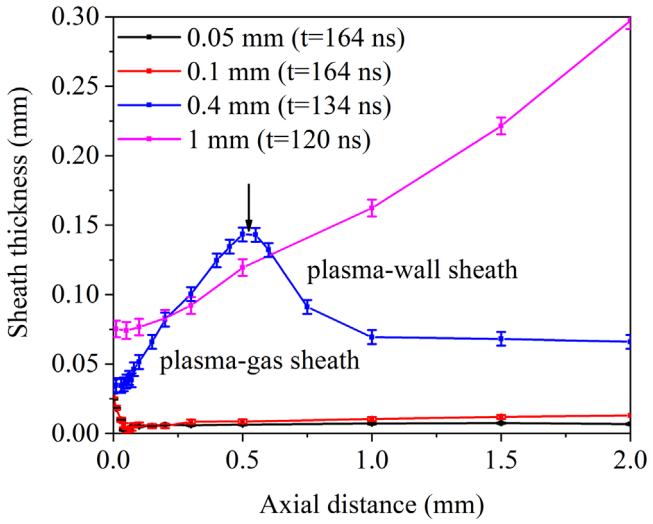


Figure 9. Axial distributions of the thickness of the plasma sheath for different diameters of dielectric tubes. Head of the plasma volume reaches the same axial position as that in figure 5. Axial distance of zero is the position of the head of the plasma volume. Labeled time corresponds to the rise in the voltage pulse.

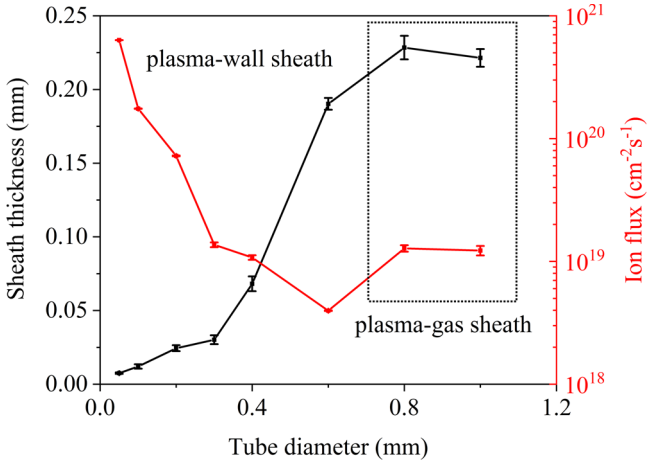


Figure 10. Curves of the sheath thickness and the ion flux versus the tube diameter at the fixed axial distance of 1.5 mm. Head of the plasma volume reaches the same axial position as that in figure 5.

In addition to the sheath thickness, ion flux is another essential parameter of the plasma sheath. The ion flux Γ_i can be estimated by [56].

$$\Gamma_i = u_B n_i = (kT_e/m_i)^{1/2} \cdot n_i = 9.8 \times 10^5 (T_e/M)^{1/2} \cdot n_i, \quad (12)$$

where u_B is the Bohm velocity, n_i is the ion density at the boundary of the plasma sheath, T_e is the electron temperature at the boundary of the plasma sheath and M is the ion mass.

Figure 10 shows the curves of the sheath thickness and the ion flux versus the tube diameter. With the increase in the tube diameter from 0.05 to 1 mm, the ion flux decreases from 6.34×10^{20} to 1.23×10^{19} cm^{-2} s^{-1}, and the sheath thickness increases from 7.47×10^{-3} to 0.228 mm. Therein, the minimum ion flux reaches 3.97×10^{18} cm^{-2} s^{-1} in the case of 0.6 mm, where the sheath transits from a plasma-wall sheath into a plasma-gas sheath. This can be explained as follows: according to [46], the relationship between the thickness of the plasma-wall sheath and the Debye length in collision-dominated plasma can be expressed by $s \propto \lambda_D^2 T_e^{1/3}$; with the growth of the tube diameter from 0.05 to 1 mm, the maximum n_e. at the axial distance of 1.5 mm decreases from 1.8×10^{15} cm^{-3} to 1.7×10^{13} cm^{-3}, and the electron temperature slightly decreases from 3.55 to 2.95 eV. Therefore, the increase in the tube diameter leads to a significant increase in the Debye length and the thickness of the plasma sheath. The ion flux is proportional to the ion density and $T_e^{1/2}$. Therefore, the increase in the tube diameter reduces the ion flux. This indicates that the plasma treatment time for improving the hydrophilicity of the polymer tubes' inner surface should be significantly shortened for small tubes.

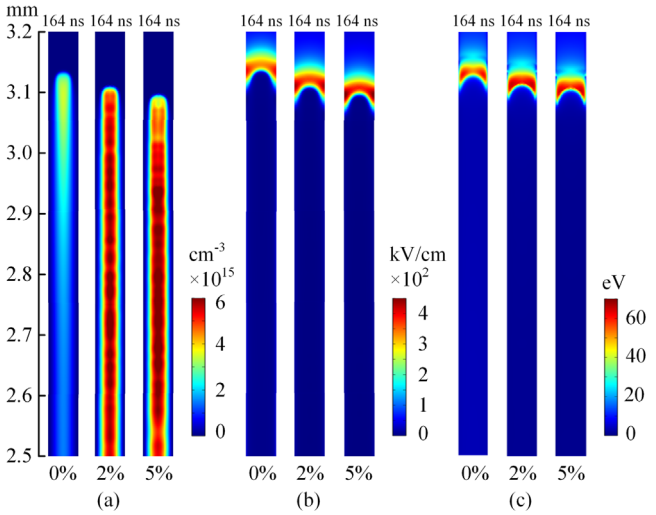


Figure 11. 2D distributions of (a) the n_e, (b) the electric field and (c) the electron temperature with different proportions of O_2 gas added to He gas. Labeled time in each graph corresponds to the rise in the voltage pulse.

3.3. Effects of O_2 addition

To improve the hydrophilicity of the polymers, noble gas mixed with O_2 gas is usually applied during APPJ treatment [7]. In addition to the electron attachment with O_2 and the Penning reactions between O_2 and He metastable, the negative ions in the plasma sheath may also impact the He discharge. To further understand how the O_2 affects the plasma sheath near the inner surface of the tube wall, numerical modeling of the He/O_2 discharge within a capillary with a diameter of 0.05 mm is implemented. Figure 11 shows the spatial distributions of the electric field, the n_e and the electron temperature with different proportions of O_2 gas. It depicts that as the O_2 gas is mixed into the He gas, both the n_e and the electron temperature increase, and the electric field in front of the head of the plasma volume is enhanced.

The curves of the n_e as a function of axial distance with different amounts of O_2 gas added into He gas are plotted in figure 12(a). The maximum n_e for the pure He is about 3.5×10^{15} cm^{-3}, significantly lower than 6×10^{15} cm^{-3} for the He/O_2(2%) gas. As the proportion of O_2 gas increases to 5%, the maximum n_e remains unchanged in the plasma head.

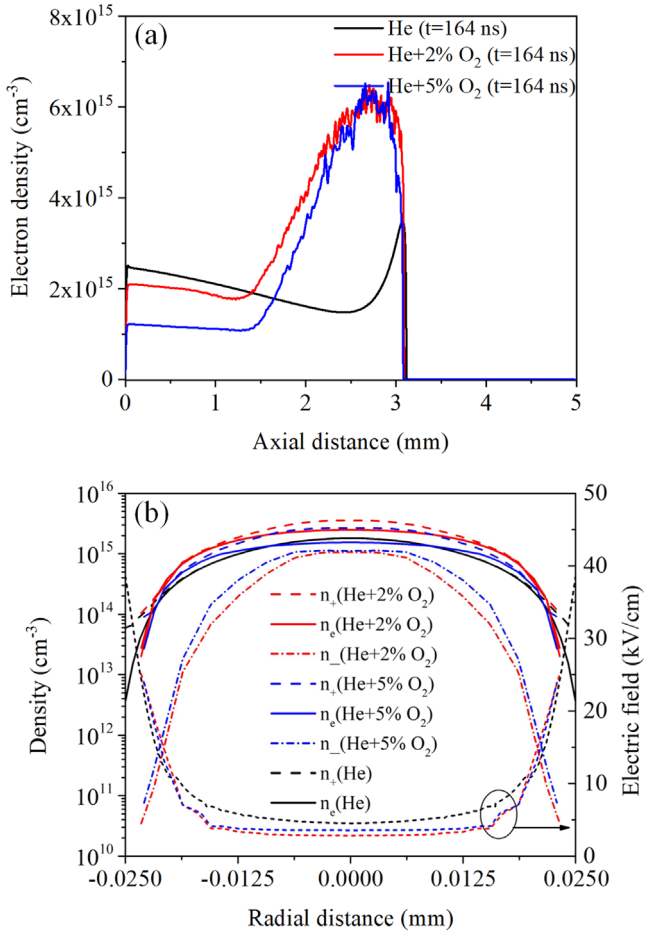


Figure 12. (a) Axial distribution of the n_e with different proportions of O_2 gas. (b) Radial distribution of the n_e, the n_+, the n_- and the electric field at an axial distance of 1.5 mm. Labeled time corresponds to the rise in the voltage pulse.

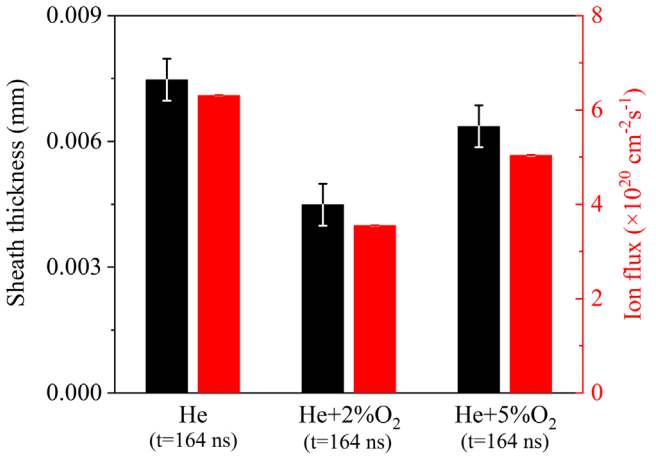


Figure 13. Thickness of the plasma sheath and the ion flux for pure He, $He/O_2(2\%)$ and $He/O_2(5\%)$ at the axial distance of 1.5 mm. Head of the plasma volume reaches the same axial position as that in figure 5. Labeled time corresponds to the rise in the voltage pulse.

In addition, for the He/O_2 gas mixture, the n_e in the plasma head is three times higher than that in the plasma column behind the plasma head.

To better understand the influence of the O_2 addition, the radial distributions of the n_e ion density, and the electric field are plotted in figure 12(b). For the He/O_2 gas mixture, the n_e is much higher than the density of the negative ions. The n_e in the plasma column in the case of the $He/O_2(2\%)$ gas mixture becomes higher than that in the case of pure He. A steep gradient of the electric field near the tube wall presents for the three cases, indicating the formation of a plasma sheath near the tube wall. The influence of the O_2 addition on the n_e can be explained as follows: because of the low ionization energy (13.6 eV) required for O_2 molecules compared to He atoms and He metastable [57], the direct ionization by the collisions between the energized electrons and O_2 molecules and the Penning ionization between He metastable and O_2 molecules contribute to the total ionization processes in the He/O_2 mixture discharge [58, 59]. Therefore, a small amount of O_2 gas increases the n_e, which is observed in the head of the plasma volume where the gas ionization processes take place. On the other hand, when the O_2 concentration further increases, the fast electron attachment by O_2 molecules is enhanced [55], which results in the depletion of the electrons and, therefore, the decrease in the n_e. This is confirmed by the reduction of the n_e in the plasma channel as the O_2 proportion increases.

Figure 13 shows the thickness of the plasma sheath and the ion flux for different proportions of O_2 gas. As the O_2 proportion increases from 0% to 5%, both the sheath thickness and the ion flux decrease at first and then increase after reaching the minimum values of 4.49×10^{-3} mm and $3.55 \times 10^{20}\,cm^{-2}\,s^{-1}$ in the case of $He/O_2(2\%)$. According to the simulation results, the electron temperature at the boundary of the plasma sheath is 3.51 eV for pure He, 2.01 eV for $He/O_2(2\%)$ and 1.95 eV for $He/O_2(5\%)$. According to the profiles of the n_e in figure 12(b), due to the sheath thickness proportional to the $\lambda_D^2 T_e^{1/3}$, a minimum sheath thickness is obtained in the case of the $He/O_2(2\%)$. On the other hand, the ion density at the boundary of the plasma sheath is $4.86 \times 10^{14}\,cm^{-3}$ for pure He, $3.59 \times 10^{14}\,cm^{-3}$ for $He/O_2(2\%)$ and $5.21 \times 10^{14}\,cm^{-3}$ for $He/O_2(5\%)$, different to the trend of the n_e at the axis. Therefore, due to the ion flux being proportional to $n_+ T_e^{1/2}$, a minimum ion flux is also obtained in the case of the $He/O_2(2\%)$.

4. Conclusion

2D numerical modeling of the microplasma propagation inside tubes is established to investigate the characteristics of the sheath between the plasma column and the inner wall. The main conclusions are obtained as follows.

(1) With the decrease in the tube diameter from 1 to 0.2 mm, both the n_e and the electric field strength significantly increase, which causes the generation of a plasma layer surrounding the central plasma column. The central plasma column disappears due to the strong field shielding effects in the case of 0.1 mm. This results in the observation of a ring-shaped mode of the He discharge, which transits into a

solid-shaped mode because of the spatial constraints of the tube wall with a diameter of 0.05 mm.

(2) Because the ions do not have enough time to drift to the inner surface of the tube during the plasma propagation, a plasma-gas sheath surrounding the central plasma column is observed for the tube diameter equal to or greater than 0.8 mm. With the further reduction of the tube diameter, the plasma-gas sheath is replaced by a plasma-wall sheath. With the growth of the tube diameter, the thickness of the plasma-wall sheath increases, and the ion flux decreases significantly. This indicates that the treatment time for the plasma modification should be largely reduced for small polymer tubes.

(3) As the O_2 gas admixture with low concentration is added to He gas in the case of 0.05 mm, both the n_e in the head of the plasma volume and the electric field in front of the plasma head increase, while the electron temperature decreases. As the O_2 proportion grows from 0% to 5%, both the thickness of the plasma sheath and the ion flux decrease at first and then increase slightly after reaching minimum values for the O_2 proportion of 2%. This implies that, with the addition of a small amount of O_2, the plasma etching effects on the surface of the inner wall may be weakened, while the generation of the O_2-related functional groups for the surface modification is probably improved.

This study provides some insight into the plasma sheath in He capillary discharge and the influence of the tube diameter and the addition of O_2 on it, which is beneficial to the plasma modification of polymer tubes.

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